

Status Report #6

2025-10-29

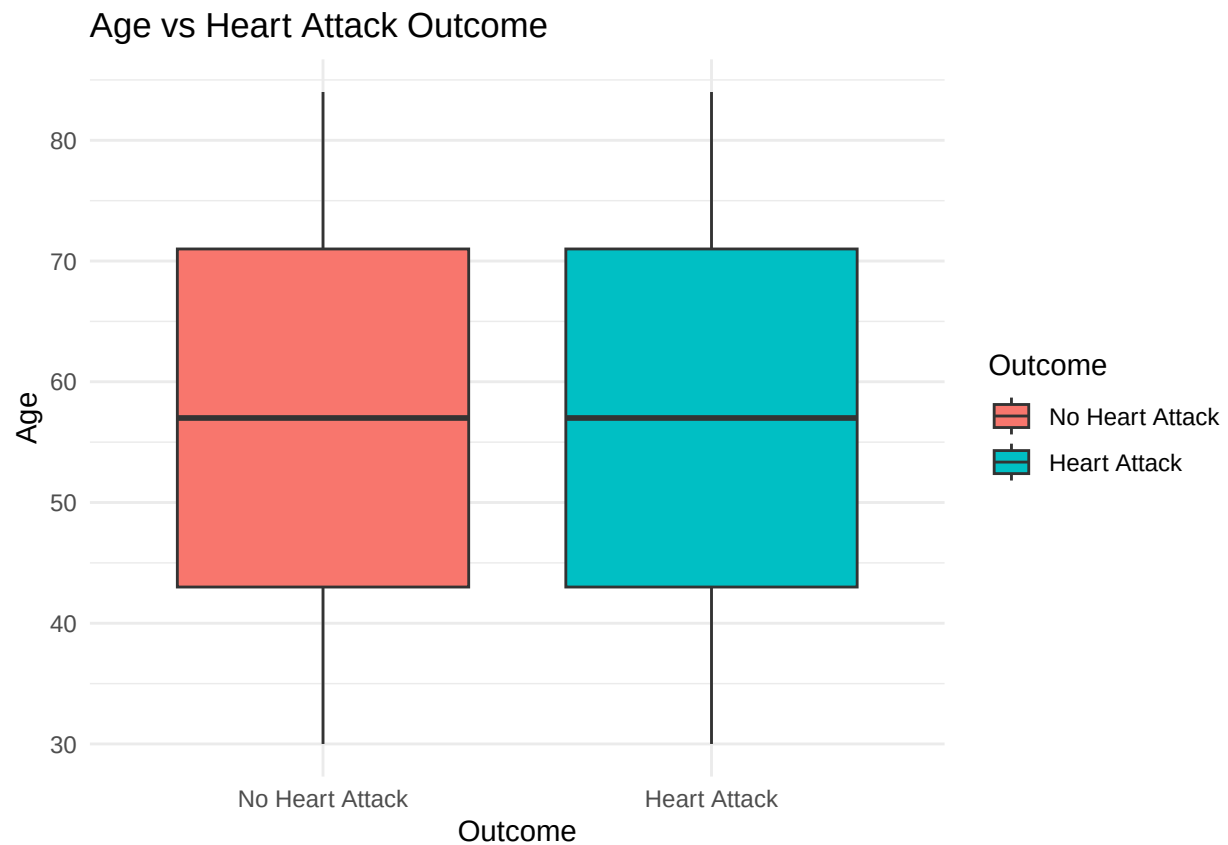
```
# Load cleaned data from Status Report #1 and view it (make sure it looks right)
HeartAttackClean <- readRDS("HeartAttackClean.rds")

# Load libraries
library(ggplot2)
library(dplyr)

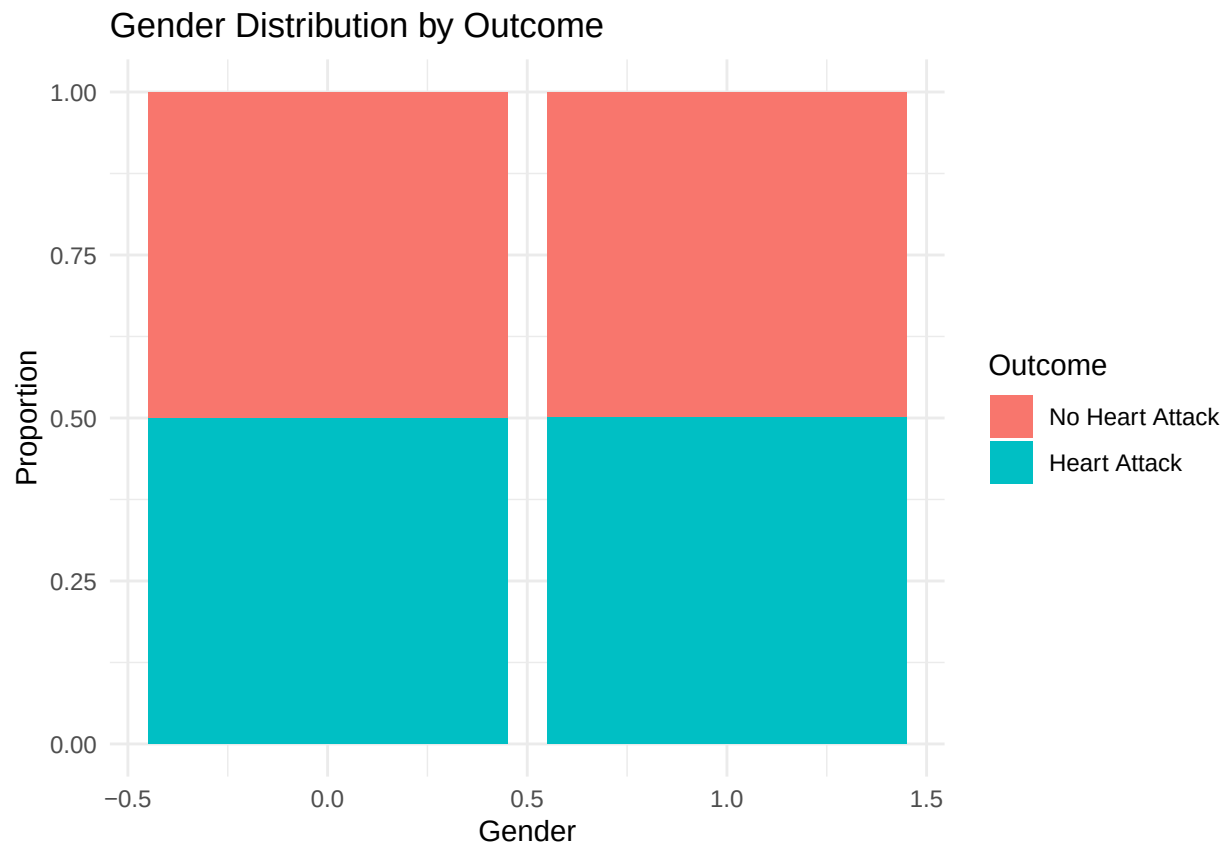
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

# Convert Outcome to a labeled factor
HeartAttackClean <- HeartAttackClean %>%
  mutate(Outcome = factor(Outcome,
                           levels = c(0, 1),
                           labels = c("No Heart Attack", "Heart Attack")))

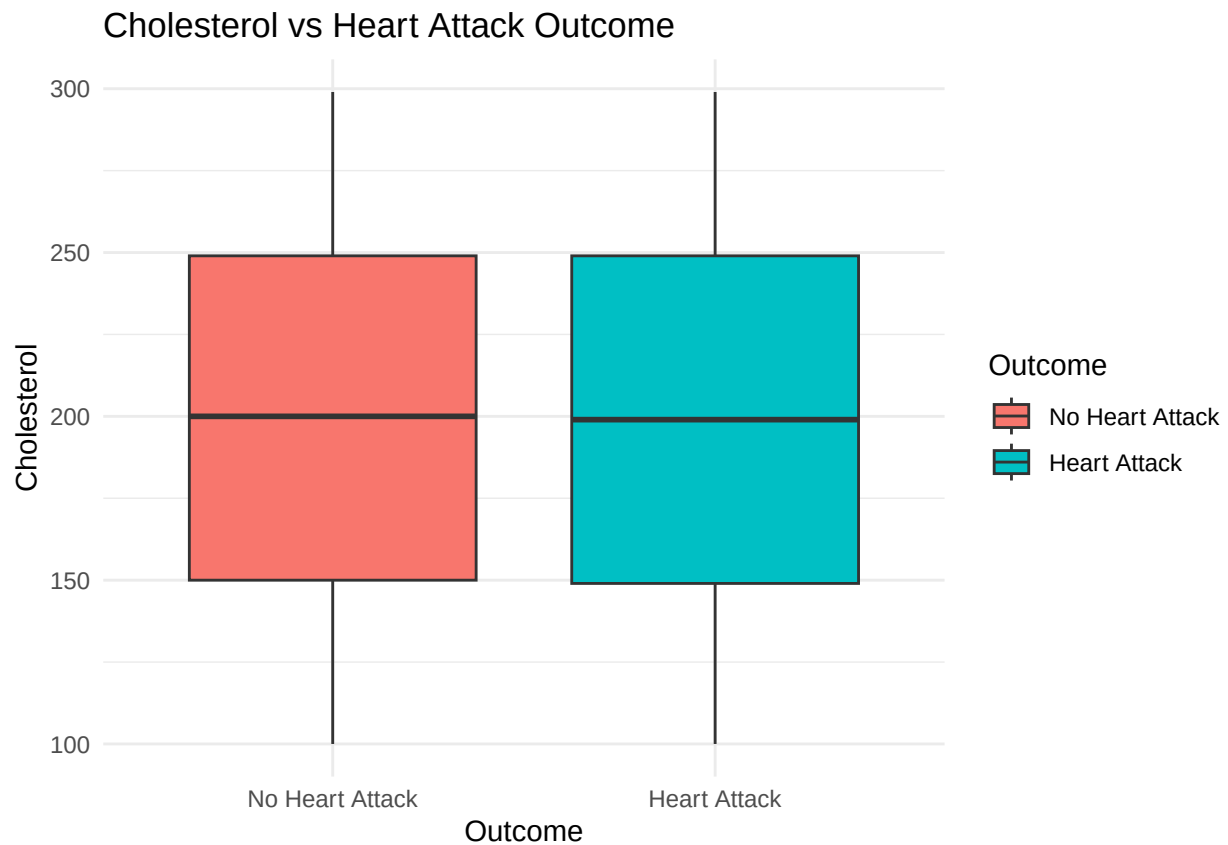
# Age vs Outcome (boxplot)
ggplot(HeartAttackClean, aes(x = Outcome, y = Age, fill = Outcome)) +
  geom_boxplot() +
  labs(title = "Age vs Heart Attack Outcome",
       x = "Outcome", y = "Age") +
  theme_minimal()
```



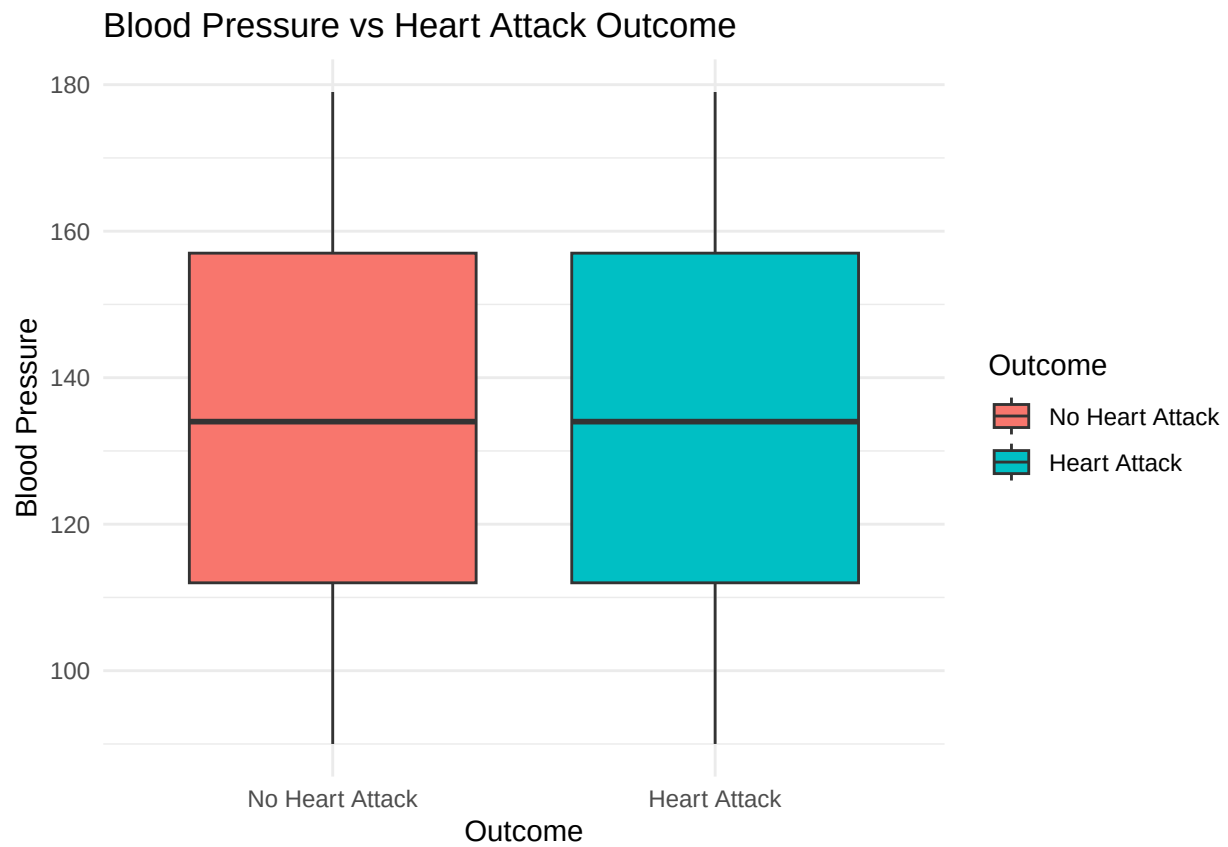
```
# Gender vs Outcome (bar chart)
ggplot(HeartAttackClean, aes(x = Gender, fill = Outcome)) +
  geom_bar(position = "fill") +
  labs(title = "Gender Distribution by Outcome",
       x = "Gender", y = "Proportion") +
  theme_minimal()
```



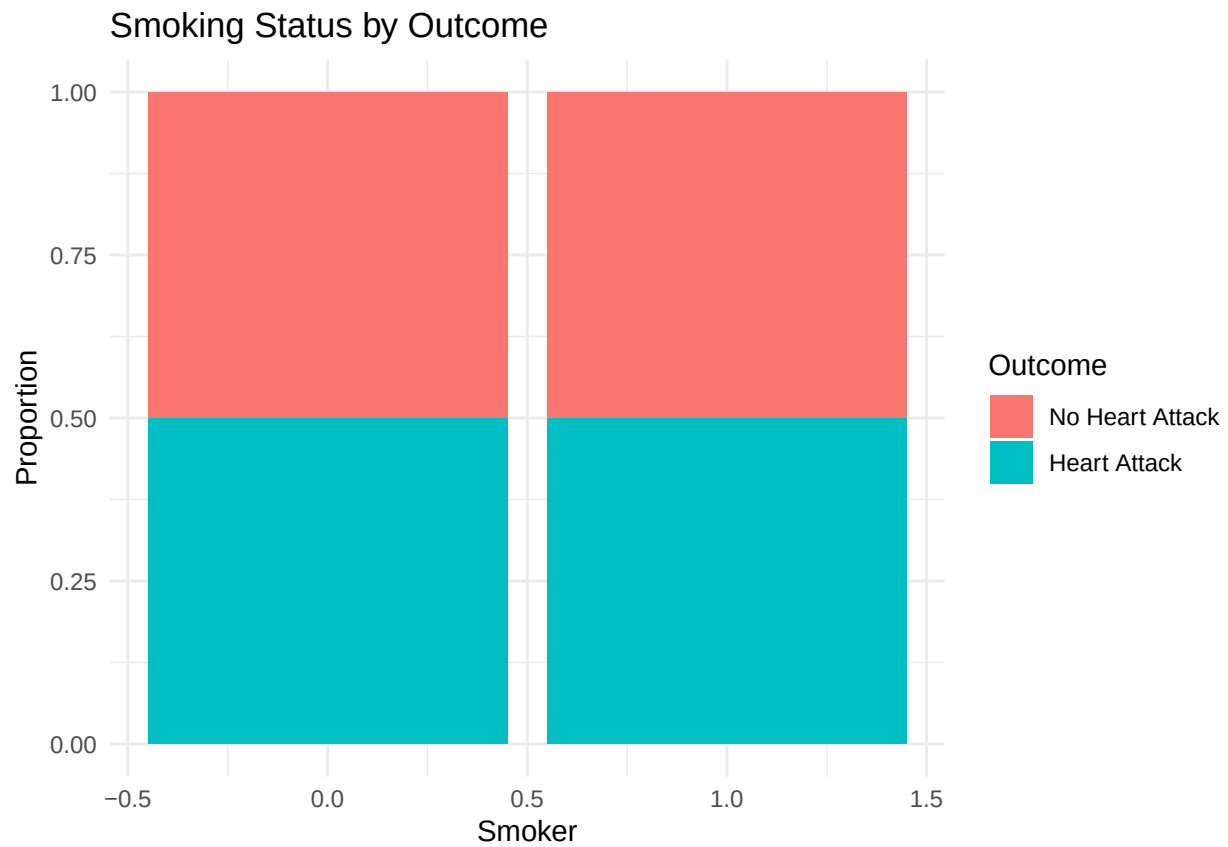
```
# Cholesterol vs Outcome (boxplot)  
ggplot(HeartAttackClean, aes(x = Outcome, y = Cholesterol, fill = Outcome)) +  
  geom_boxplot() +  
  labs(title = "Cholesterol vs Heart Attack Outcome",  
        x = "Outcome", y = "Cholesterol") +  
  theme_minimal()
```



```
# Blood Pressure vs Outcome (boxplot)
ggplot(HeartAttackClean, aes(x = Outcome, y = BloodPressure, fill = Outcome)) +
  geom_boxplot() +
  labs(title = "Blood Pressure vs Heart Attack Outcome",
        x = "Outcome", y = "Blood Pressure") +
  theme_minimal()
```



```
# Smoking vs Outcome (bar chart)
ggplot(HeartAttackClean, aes(x = Smoker, fill = Outcome)) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status by Outcome",
       x = "Smoker", y = "Proportion") +
  theme_minimal()
```



```
# Physical Activity vs Outcome (bar chart)  
ggplot(HeartAttackClean, aes(x = PhysicalActivity, fill = Outcome)) +  
  geom_bar(position = "fill") +  
  labs(title = "Physical Activity by Outcome",  
        x = "Physical Activity", y = "Proportion") +  
  theme_minimal()
```

